# AI Messaging Sales Assistant

One AI assistant that answers customers on WhatsApp, Instagram and Messenger, books appointments, captures leads, and hands over to staff. Built for any business, with self-service sign-up.

Planning document, version 5 (concise)

## 1. Goal

Answer every customer message within seconds, 24 hours a day, in Chinese and English. Capture leads, book appointments, and hand over to a person whenever one is needed.

| For | What it delivers |
|---|---|
| Customer | Instant replies in the chat app they already use, in their own language. A person is always one message away. |
| Staff | One inbox for all channels, an AI summary on every handover, and a simple way to give the chat back to the AI. |
| Business owner | A web admin panel to change prices, FAQ and AI behaviour without a developer, plus a daily summary on the phone. |
| Platform owner (you) | Businesses sign up and set up by themselves. You see usage and revenue, and can pause any business. |

## 2. How it works

## 3. What the system includes

| Module | What it does |
|---|---|
| Chat | One inbox for three channels. AI answers first, staff take over at any time. Assignment, labels, private notes, canned replies. |
| Customers | Searchable list with filters, tags, tiers and saved views. One customer ID across channels (matched by phone). Full chat and booking history. |

| Module | What it does |
|---|---|
| Bookings | Real availability from Google Calendar, **one calendar per practitioner or resource**. Book, reschedule, cancel, reminders. |
| Marketing | Template broadcasts and reactivation, only to customers who opted in. |
| Workflow | Automations (routing, reminders, follow-ups) edited visually in n8n. |
| Reports | Chats, leads by score, bookings, handovers, reply time, top questions. |
| Team | Staff accounts, roles, automatic assignment, routing by skill. |
| Settings | AI behaviour, catalog, resources, FAQ, hours, alerts, channel connections, plan and billing. |

### Smart extras

- **AI suggested replies** for staff, sent with one click.
- **Auto summary, tags, intent and mood** on every chat.
- **Translation view** so staff read and reply across Chinese and English.
- **Click-to-chat link and QR code** that also records where each customer came from.
- **After-hours mode**, **self-service rescheduling** in chat, and a **weekly owner insights** report.
- Later: aftercare and review requests, rebooking reminders, no-show recovery, waiting list, knowledge import from a price list or web page, review queue for unsure answers, deposit links.

## 4. For every industry, with self-service sign-up

| Building block | How it fits any business |
|---|---|
| Universal base | Replies, FAQ answers, lead capture and scoring, handover. Works even with no booking. |
| Catalog | Services or products with name, price and description in each language. |
| Booking types | Appointment with a resource, table, viewing, class, quote visit, or none. |
| Custom lead questions | The owner chooses what to ask (budget, party size, location and so on). |
| Starter templates | Clinic, dental, salon, real estate, restaurant, education, repair, retail and more. Pre-fill everything, fully editable. |
| AI-assisted setup | Drafts the setup from a description, website or price list. The owner approves before going live. |

### Sign-up journey

| | |
|---|---|
| 1 | Sign up and verify email |
| 2 | Describe the business, or paste the website or price list |
| 3 | AI drafts the setup, owner reviews and approves |
| 4 | Connect Google Calendars and WhatsApp with guided steps and test buttons |
| 5 | Test chat, choose a plan, go live |

## 5. Parts and tools

| Part | Tool | Role |
|---|---|---|
| Channels | Meta WhatsApp, Instagram, Messenger APIs | Where customers write. Each business uses its own number and Meta account. |
| Inbox | Chatwoot (open source) | Staff inbox. One account per business, created automatically. |
| Automation | n8n (open source) | Routes messages, sends replies, handover alerts, reminders, reports. |

| Part | Tool | Role |
|---|---|---|
| AI service | Python service in the aichat repo + Claude | Prompt, business knowledge and tools. Replies in the customer's language. |
| Admin panel | Web pages in the AI service | Business settings, customers, bookings, reports. Platform-owner console. |
| Calendar | Google Calendar API | Availability and bookings, one calendar per practitioner. |
| Database | PostgreSQL | All data, separated per business with row-level rules. |
| Sign-up and billing | Public site + Stripe or local provider | Registration, plans, usage limits, invoices. |

## 6. Handover, safety and separation

### The AI hands over to staff when

- The customer asks for a person.
- Complaint, anger, refund or payment problem.
- Medical, legal or safety concern. The AI never gives such advice.
- The question is outside the business's knowledge, or the AI is unsure.

### Safety and separation

- Answers only from the business's own knowledge. Never invents prices, offers or availability.
- Tells customers it is a virtual assistant. Customer text is treated as untrusted.
- Every record carries a business ID, and database rules block cross-business reads.
- Encrypted credentials, two-step login for owners, audit logs, data export and deletion.
- Sign-up protection: email verification, bot check, limits, and a kill switch per business.
- If the AI service fails, the customer gets a polite message and staff are alerted.

## 7. Plans, hosting and costs

| Item | Approach |
|---|---|
| Plans | Free trial, then paid monthly plans with limits on AI replies, staff and resources. At the limit the AI pauses, the owner is told, and messages still reach the inbox. |
| Hosting | Free hosting for development, the demo and an invite-only beta. Public sign-up needs a stable paid server with backups. |
| AI cost | Pay per use and small (Claude Haiku 5.5 by default). Counted per business and passed on through plans. |
| Meta fees | Billed by Meta directly to each business. |
| Other | Domain, email sending, and payment provider fees. |

## 8. Build order

| Step | What gets built |
|---|---|
| 1 | Accounts and Meta approvals (start first, they can be slow). Docker stack with multi-business base. |
| 2 | Core bot on WhatsApp: bilingual answers, leads, handover. |
| 3 | Booking with Google Calendar, reminders. |
| 4 | Business admin panel, customers module. |
| 5 | Self-service sign-up, AI-assisted setup, plans and billing. |

| Step | What gets built |
|---|---|
| 6 | Platform-owner console, safety, legal pages. Instagram and Messenger. |
| 7 | Reports, hardening, smart extras, marketing and workflows. |

Recommended path: run an invite-only beta first, then open public sign-up once Meta approvals and a paid server are in place.

## 9. Main risks

| Risk | Reduction |
|---|---|
| Meta approvals delay launch | Start them first and run the beta meanwhile. |
| AI gives a wrong answer | Answer only from approved knowledge, handover rules, test chat, review queue. |
| Abuse or spam sign-ups | Verification, limits, caps, kill switch. |
| One business affecting another | Business ID everywhere, row-level rules, separation tests, own Meta account per business. |
| Legal and privacy duties | Terms, privacy policy, data export and deletion. Have a lawyer review for your country. |

## 10. Decisions needed

- Platform name, domain, and the country billing starts in.
- Plan structure: trial length, limits, prices, payment provider.
- The company that owns the Meta business verification and signs the terms.
- Start with an invite-only beta, then open sign-up.
- Where it runs for now: free cloud VM (recommended) or an office machine.